

1. The Goal

Find a target value in a matrix where every row is sorted, and the start of a row is greater than the end of the previous row.

2. The Intuition

Since the end of Row 0 connects to the start of Row 1 perfectly, this is just one long sorted array that has been cut into pieces. Do not search row-by-row. Search the whole thing at once.

3. The Math (Index Mapping)

Given a matrix with 'C' columns:

- To get 1D index from 2D: $\text{index} = r * C + c$

- To get 2D Coords from 1D:

$$\rightarrow \text{row} = \text{index} // C$$

$$\rightarrow \text{col} = \text{index} \% C$$

4. Visual Trace

matrix (3x4):

[1, 3, 5, 7]

target = 16

[10, 11, 16, 20]

[23, 30, 34, 60]

1. Range: [0, 11] (12 elements total)

2. mid: $(0+11) // 2 \Rightarrow \underline{5}$

3. map: Index 5 $\rightarrow 5 // 4 = 1$ (Row 1), $5 \% 4 = 1$ (Col 1)

4. Value: matrix[1][1] is 11.

5. Compare: $11 < 16$. We need higher numbers

6. New Range: start becomes mid+1 (6). Range [6, 11]

7. New mid: $(6+11) // 2 \Rightarrow \underline{8}$

8. map: Index 8 $\rightarrow 8 // 4 \Rightarrow 2$ (Row 2), $8 \% 4 = 0$ (Col 0)

9. Value: matrix[2][0] is 23.

10. Compare: $23 > 16$. We need lower numbers.

11. New Range: end becomes mid-1 (7). Range [6, 7] ... and so on...

Find the Row (how many lines down?): You have passed 4 characters on the first line. $6 // 4 = 1$ (with remainder).

This means you have completed 1 full line and are now on the next one (Row 1).

Find the Column (how far across the current line?):

After filling the first line (4 characters), how many are left over? $6 - 4 = 2$ ($6 \% 4 = 2$) (Remainder). You are at position 2 on the current line.